

Week 4 : Assignment 4

Due date on 2026-08-19, 23:59 IST

The due date for submitting this assignment has passed. Your last recorded submission was on 2026-08-18, 22:10 IST.

1. A context switch occurs when a state of process changes from new to ready

1 Point

- ☐ True
- ☒ False

Yes, the answer is correct. Score: 1

Accepted answer is False

2. A thermal sensor in the CPU checks the temperature of the machine. If it is more than 40°C an interrupt is generated to shutdown the machine. This interrupt is a hardware interrupt.

1 Point

- ☒ True
- ☐ False

Yes, the answer is correct. Score: 1

Accepted answer is True

3. What is the correct order of the following events

1 Point

- I. device raises interrupt
II. CPU checks signal on INT pin and sends INT ACK signal
III. Interrupt Handler Routine is executed
IV. Interrupt Controller sends INT signal to CPU
V. CPU identifies which device has raised the interrupt

- ☐ I,II,III,IV,V
- ☐ V,I,IV,II,III
- ☒ I,IV,II,V,III
- ☐ II,IV,I,V,III

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1 Point

- I. device raises interrupt
II. CPU checks signal on INT pin and sends INT ACK signal
III. Interrupt Handler Routine is executed
IV. Interrupt Controller sends INT signal to CPU
V. CPU identifies which device has raised the interrupt

- ☐ I,II,III,IV,V
- ☐ V,I,IV,II,III
- ☒ I,IV,II,V,III
- ☐ II,IV,I,V,III

Yes, the answer is correct. Score: 1

Accepted answer is I,IV,II,V,III

4. I/O APICs are used to handle interrupt from thermal sensors and internal timers

1 Point

- ☐ True
- ☒ False

Yes, the answer is correct. Score: 1

Accepted answer is False

5. Match the following Mnemonics of Interrupt Vector used in x86 to the type of interrupt they are

1 Point

- I. #0F a. Trap
II. #DF b. Fault
III. #SS c. Abort

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5. Match the following Mnemonics of Interrupt Vector used in x86 to the type of interrupt the are

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- I. #0F a. Trap
II. #DF b. Fault
III. #SS c. Abort

- ☒ I - a II - b III - c
☐ I - a II - c III - b
☐ I - b II - c III - a
☐ I - c II - b III - a

No, the answer is incorrect. Score: 0

Accepted answer is I - a II - c III - b

6. A CPU with four processes executing has an Operating System with a non-preemptive scheduler. What is the total number of scheduling possible for the 4 processes?

1 Point

24

Yes, the answer is correct. Score: 1

Accepted answer is Type: Numeric: 24

7. More than one System call in an operating system can have the same system call number

1 Point

- ☐ True
☒ False

Yes, the answer is correct. Score: 1

Accepted answer is False

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8. Parameters in system calls are passed using user mode stacks in Linux and by using registers in xv6.

1 Point

- ☐ True
☒ False

Yes, the answer is correct. Score: 1

Accepted answer is False

9. Read the following statements and point out which are true and false

1 Point

- I. A context Switch can occur when a process moves from a blocked to ready state
II. A context Switch can occur when a process moves from a running to zombie state
III. A context Switch can occur when a process moves from a running to ready state

- ☐ I - False, II - True, III - True
☐ I - False, II - False, III - True
☒ I - True, II - True, III - True
☐ I - False, II - True, III - False

No, the answer is incorrect. Score: 0

Accepted answer is I - False, II - True, III - True

10. Timer Interrupt Latency should be as high as possible for smooth context switching

1 Point

- ☐ True
☒ False

Yes, the answer is correct. Score: 1

Accepted answer is False